

TUSHAR KUMAR

+91 9917677832 tusharparthsarathi@gmail.com
[linkedin.com/in/tusharkumarparthsarathi](https://www.linkedin.com/in/tusharkumarparthsarathi) github.com/VincitoreSi

Professional Summary

AI Engineer with 2+ years building production AI/ML systems and scalable applications across enterprise and research environments. Currently **Software Engineer I** at **Samsung R&D India** contributing to large-scale **Tizen enterprise systems** and AI-powered developer tools (**Good Idea Award** winner). Built multi-agent GenAI pipelines, ML applications with PyTorch, and end-to-end AI systems including podcast generation serving **5,000+** students. IIT Jodhpur alumnus (**B.Tech. AI & Data Science**).

Experience

Samsung R&D Institute India, Delhi

Software Engineer I

Jul 2024– Present

Noida, UP, India

Tizen Signage Core Modules

- Managed Tizen modules (SignagePlayer, OfficeViewer, Event Manager, Remote Manager) across **100+** enterprise deployments.
- Reduced memory usage by **25%** via profiling and C/C++ optimization; resolved **50+** production crashes.
- Implemented subtitle functionality in SignagePlayer, enhancing accessibility for global customers.

Smart Diagnostics & Logging

- Enhanced diagnostics collecting logs, CPU, and memory telemetry across **10,000+** Samsung devices.
- Cut issue resolution time by **30–40%**; scaled log collection to **2GB+** per session.
- Integrated logging pipeline into org-wide portal, enabling access for **500+** engineers across 20+ countries.

AI for Education & Developer Productivity

- Engineered AI podcast generation system producing **100+** hours of lecture content for **5,000+** students.
- Architected Code2Documentation POC – won Samsung **“Good Idea Award”** (1 of 50 globally) with cash prize.

Education

Indian Institute of Technology (IIT) Jodhpur

B.Tech. Artificial Intelligence & Data Science

Jul 2020 – May 2024

Technical Skills

Languages: C/C++, Python, Java, JavaScript/TypeScript, Go, SQL

Frameworks: PyTorch, TensorFlow, Pandas, NumPy, Scikit-learn, FastAPI, Django, Flask

Tools & OS: Git, GitHub, Docker, AWS, Linux, Neovim, L^AT_EX, Kaggle

Web & IoT: ReactJS, NextJS, NodeJS, HTML/CSS/JS, Raspberry Pi

Projects

The Cutting Room | FastAPI, React 19, Docker, Gemini, ffmpeg, Python/TypeScript

2026

- Built multi-agent content virality pipeline with FastAPI hub + React 19 dashboard; 4 agents communicate solely over HTTP with no shared filesystem.
- Engineered virality scoring engine blending 4 percentile-normalized engagement signals (rate, reach, outlier, velocity) into 0–100 score with configurable tier gate.
- Containerized as multi-arch Docker image with custom ~130MB-minified ffmpeg; shipped with 4 CI/CD workflows and published to GHCR.
- Implemented Gemini 2.5 Pro blueprint generation with jsonschema validation + automatic repair loops for shot-by-shot content breakdowns.

Electronic Nose System for Gas Sensor Data | Python, PyTorch, Raspberry Pi

Sep 2023 – Apr 2024

- Designed end-to-end ML pipeline processing **10,000+** sensor readings for gas classification and concentration prediction.
- Collected real-time data via Raspberry Pi with ThingSpeak integration; built interactive monitoring dashboard.

Neural Tangent Kernel Aware Pruning in DNN | PyTorch

Jan 2023 – May 2023

- Proposed novel pruning method based on Neural Tangent Kernel (NTK) theory; reduced model connections while preserving training dynamics.
- Achieved ~20% model compression with minimal accuracy loss; improved computational efficiency.
- Conducted under Prof. Binod Kumar at IIT Jodhpur as part of academic research on efficient deep learning.

Handmade Sketch Colorization using Segmentation | PyTorch, Docker, Streamlit

Jan 2022 – May 2022

- Shipped ML-based web application for automated sketch colorization using image segmentation.
- Containerized pipeline with Docker; deployed via Streamlit supporting **100+** concurrent sessions.